

Analytical Review of Graph-Conditioned Neural Decompilation of Dart AOT Binaries

Executive summary

This manuscript is best understood as an **empirical machine-learning systems paper** aimed at readers in reverse engineering, program analysis, neural code generation, and software engineering research. Its present structure and numbered citation style make it look far closer to an **IEEE/ACM-style technical paper** than to an APA-style manuscript. If the intended venue is double-blind, however, the current title page and self-citations would need immediate anonymisation; if the intended venue is a journal with a house style, that guide should take precedence over any generic convention.

filecite turn0file 1

Overall, the paper has a **promising core contribution**: it identifies a genuinely important phenomenon in neural decompilation, namely that better surface metrics do not necessarily yield better functional correctness, and it presents a useful empirical contrast between graph conditioning, reward adaptation, reranking, and multi-arm candidate pooling. The paper is also sensibly organised at a high level, with Introduction, Related Work, Method, Experimental Setup, Results, Discussion, Threats to Validity, Future Work, and Conclusion. filecite turn0file

My overall recommendation is **major revision**. The key reasons are not that the paper lacks merit, but that several issues currently weaken its scientific reliability and publication readiness:

Area	Main issue	Why it changes the recommendation
Experimental validity	The paper openly acknowledges that some training and evaluation arms share data and that harvest/ensemble results should be treated as optimisation analysis rather than clean generalisation, yet the clean and transductive-style results are still presented too closely together. filecite turn0file	This threatens the clarity of the central claim. Readers need a sharper boundary between generalisable evidence and benchmark-specific candidate harvesting .
Metrics and statistical reporting	Section 4 says the paper reports compile@1 and compile@5, but Table 1 shows only compile@5 and a redundant “Compiled” count. There are also no confidence intervals, no significance tests, and no explicit clarification of whether pass@k is an empirical k-sample success rate or an estimator in the HumanEval sense. filecite turn0file 2	This makes comparisons harder to interpret and reduces comparability with prior code-generation literature.

Area	Main issue	Why it changes the recommendation
Reproducibility	There is no explicit code/data/checkpoint availability statement, no seeds, no sampling hyperparameters, no GRPO training details sufficient for replication, and no appendix listing exact run configurations beyond partial mentions in prose. filecite turns file	For an empirical ML paper, this is a major weakness.
Readability and style	The manuscript relies heavily on internal run labels such as <i>G3</i> , <i>p128r</i> , <i>withB ref ultralite</i> , and <i>binary repair ultralite</i> , which are meaningful to the authors but not yet reader-friendly. filecite turns file	This obscures the argument and forces the reader to decode the experiment taxonomy while trying to follow the results.
Reference hygiene	The references are inconsistent: some entries include venue and pages, many do not; several likely arXiv or technical-report items omit persistent identifiers; and one reference is explicitly “under peer review”, which is weak evidence for readers and potentially problematic for anonymous review. filecite turns file 3	A strong paper in this area needs a cleaner, more complete reference apparatus.

Bottom line: the paper appears publishable in principle, but not in its current state. The right path is **major revision with re-framing**, not rejection by default. The most important improvement is to restructure the evidence so that the reader can clearly distinguish **clean model comparison**, **reranking analysis**, and **candidate-harvesting/coverage analysis**. [filecite turns file](#)

Assumptions and review frame

Because the paper did not specify its target audience, venue, or paper type in metadata, this review proceeds under explicit assumptions derived from the manuscript itself.

Item	Working assumption	Basis in the manuscript
Paper type	Empirical benchmarking / systems paper rather than theoretical or survey paper	The manuscript presents datasets, architecture variants, experimental arms, tables of metrics, and a threats-to-validity section rather than proofs or a literature synthesis as its main output. filecite turns file
Likely audience	Researchers in reverse engineering, ML for code, and software engineering	The paper foregrounds decompilation, CodeBLEU, pass@k, HumanEval-Dart, GraphCodeBERT, LoRA, and Qwen. filecite turns file

Item	Working assumption	Basis in the manuscript
Closest default style	IEEE/ACM technical style	The paper uses numbered references, bracketed citations, and conventional technical sectioning; IEEE's author manual expects a one-paragraph abstract, index terms, and a standard article order that is broadly similar to what the paper already uses. filecite turn0file0 ⁴
APA fit	Low unless the manuscript is completely reformatted	APA is an author-date system and is most natural in social/behavioural sciences rather than a systems-style SE/ML paper of this form. ⁵
Elsevier fit	Possible, but journal-specific	This cannot be assessed properly until a target journal is named; for Elsevier, the specific Guide for Authors would govern the final form.

This matters because several apparent “problems” are in fact **venue-dependent**. For example, the presence of author names is normal for a camera-ready or non-blind draft, but not for an anonymous review submission. Likewise, “Figure 1” may be acceptable in a generic preprint, whereas IEEE style expects figure and table citation conventions to be handled in a particular way. [filecite turn0file0](#)

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Findings across the requested dimensions

Dimension	What is working	Errors, inconsistencies, and style issues	Why they matter	Priority correction
Overall structure and organisation	The macro-structure is strong: title, abstract, Introduction, Related Work, Method, Experimental Setup, Results, Discussion, Threats to Validity, Future Work, Conclusion, References. The abstract is one paragraph and is approximately within the IEEE 150–250 word range. filecite turn0file0 ⁴	Missing keywords/index terms ; missing author affiliations/contact details if this is not blind; no code/data availability statement ; no explicit ethics/broader-impact note; no short roadmap of contributions at the end of the Introduction. filecite turn0file0	These omissions reduce discoverability, publication readiness, and reader orientation. IEEE explicitly expects Index Terms after the abstract. ⁷	Add affiliations or anonymise fully depending on venue; add Index Terms; add a concise contributions paragraph; add availability and ethics statements.

Dimension	What is working	Errors, inconsistencies, and style issues	Why they matter	Priority correction
Argumentation and logical coherence	The central narrative is sensible: sequence-only fine-tuning is insufficient, graph conditioning helps, reward tuning is unstable, ensembling yields complementary gains. filecite turn0file	The paper treats clean model comparison, reranking effects, and multi-arm harvesting/coverage too closely together. The strongest claims increasingly rely on pooled candidate coverage, but the manuscript does not elevate this shift in evidential status early enough. The phrase “These results suggest that graph conditioning and multi-policy candidate ensembling are more reliable directions...” in the abstract is slightly stronger than the evidence base warrants, especially because much of the coverage story is benchmark-specific pooling rather than clean generalisation. filecite turn0file	Readers need to know which claims are about deployable models, which are about search-space discovery, and which are about benchmark optimisation. Without that separation, the paper sounds more conclusive than it is.	Split claims into: primary clean results, secondary reranking analysis, tertiary pooled-coverage analysis. Add explicit research questions or hypotheses.

Dimension	What is working	Errors, inconsistencies, and style issues	Why they matter	Priority correction
Technical accuracy	The paper does a good job of separating the 126-task compile/CodeBLEU denominator from the 154-task pass-set denominator, and the threats-to-validity section is more candid than average. filecite turn0file0	Section 4 says the paper reports compile@1 and compile@5, but Table 1 omits compile@1. Table 1's "Compiled" column is numerically redundant with compile@5 because both represent "at least one compilable candidate among five" on the 126-task set. The paper also does not state clearly whether pass@k is a simple empirical success fraction from k samples or a HumanEval-style estimator. HumanEval's original paper formalised pass@k as a functional metric and discusses estimation from multiple samples. filecite turn0file0	Metric ambiguity reduces comparability and makes the contribution look less rigorous than it may actually be. Missing compile@1 is a direct internal inconsistency.	Restore compile@1, define pass@k exactly, state sample count and estimator formula, and add confidence intervals or paired statistical tests.

Dimension	What is working	Errors, inconsistencies, and style issues	Why they matter	Priority correction
Data presentation	The figure and tables are genuinely informative, especially in showing the trade-off between CodeBLEU, compile@k, and pass@k. filecite turn0file0	Table 1 uses cryptic labels; its caption does not explain every abbreviation; best values are not visually marked; no uncertainty intervals are shown; Table 2 headings are heavily abbreviated; Figure 1 caption is serviceable but could be more self-contained and venue-conformant. IEEE expects clear figure/table citation order and stable caption conventions. filecite turn0file0	Dense result tables increase cognitive load and slow down review. In a competitive paper, readers should not have to reverse-engineer run names from prose.	Replace internal labels with descriptive names in the main table, move archive/run IDs to an appendix, add CIs, and make captions standalone.

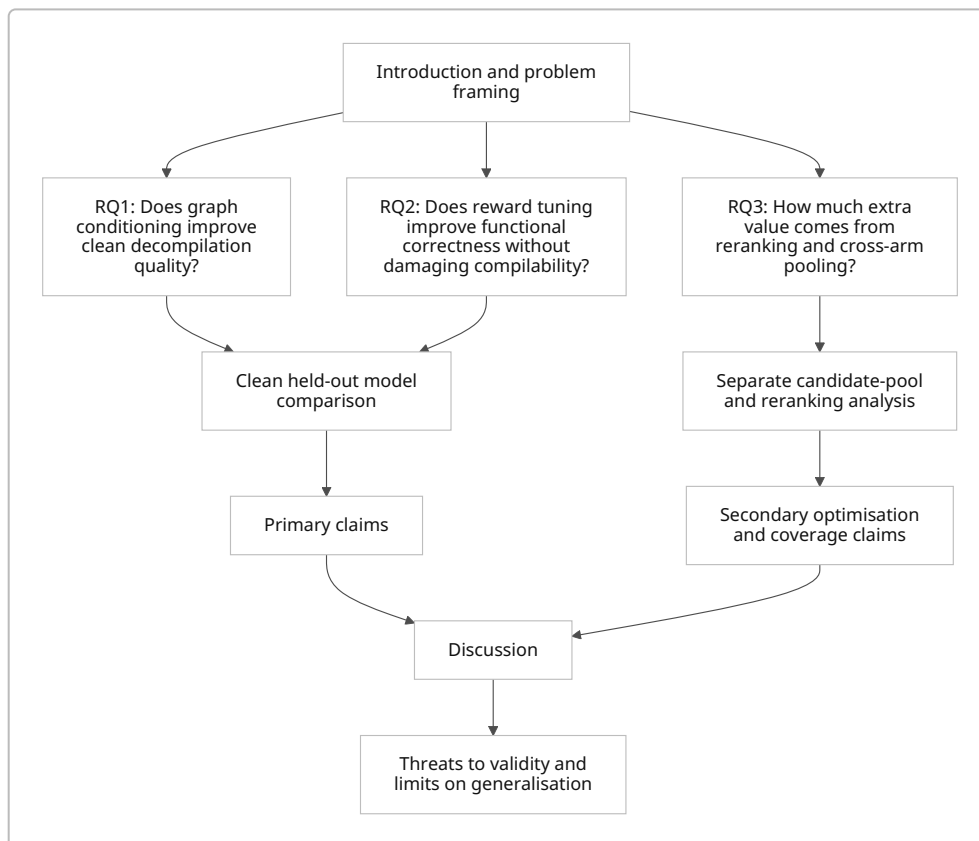
Dimension	What is working	Errors, inconsistencies, and style issues	Why they matter	Priority correction
Citations and references	The paper cites relevant prior work in decompilation, code modelling, and optimisation. filecite turn0file	Reference formatting is inconsistent. Some entries include venue and pages; many list only title and year. Several entries that are clearly available as arXiv or technical reports omit persistent identifiers, including GraphCodeBERT, LoRA, HumanEval/ Codex, DeepSeekMath, SimKO, and the Qwen3 Technical Report. The paper also includes a self-reference marked “under peer review”, which is weakly verifiable and may be problematic in blind review. IEEE expects each reference to be a separate numbered entry; Crossref and current DOI practice favour actionable DOI URLs when available. filecite turn0file	Incomplete references make it harder for reviewers to verify claims and reduce the paper’s professional polish.	Normalise all references to a single venue style; add missing venues, pages, publishers, arXiv IDs, and DOIs/URLs where applicable; reconsider how unpublished self-citations are handled.

Dimension	What is working	Errors, inconsistencies, and style issues	Why they matter	Priority correction
Language, grammar, and academic style	The prose is generally clear, and the paper avoids most obvious grammatical problems. The tone is often confident without being opaque. filecite turn0file0	Style is uneven. Informal phrases such as “one magical checkpoint” and archive-facing labels such as “withB ref ultralite” weaken academic tone. Several sentences are longer than needed and bury the main claim. The paper sometimes slides from precise description into promotional language such as “best balanced”, “strongest result”, and “substantially more” without immediate statistical framing. filecite turn0file0	Reviewers tend to trust papers more when claims are precise, hedged appropriately, and separated from interpretation.	Tighten sentences, replace colloquial phrasing, standardise tense, and distinguish observation from interpretation.

Dimension	What is working	Errors, inconsistencies, and style issues	Why they matter	Priority correction
Reproducibility and transparency	The method section gives a useful conceptual overview of CFG/DFG extraction, graph conditioning, and adaptation arms. filecite turn0file	Replication details are insufficient: no public repository, no checkpoint list, no seed reporting, no detailed sampling settings, no optimisation schedules, no exact GRPO reward definition beyond high-level description, no precise train/validation split description, and no appendix with full configs. Most importantly, the paper acknowledges that some harvest/ensemble results are not clean generalisation but does not reorganise the evidence around that admission. filecite turn0file	In ML systems papers, reproducibility is part of the contribution, not a nice-to-have.	Add a reproducibility appendix and a clearly labelled “clean vs transductive” protocol split.
Formatting and style-guide adherence	The abstract length and overall article order are compatible with IEEE-like expectations. filecite turn0file 4	Missing Index Terms; uncertain byline status; figure/table style not fully harmonised with IEEE norms; references not normalised; if APA were intended, the entire citation apparatus would need conversion to author-date format. filecite turn0file 10	Style mismatch can trigger desk-level friction even when the science is sound.	Choose the target venue first, then do a full style pass once.

Dimension	What is working	Errors, inconsistencies, and style issues	Why they matter	Priority correction
Originality and plagiarism concerns	No obvious plagiarism is visible from the text alone. The paper does cite its own earlier work rather than presenting those ideas as if new. fileciteurlturn0file0	There is a self-overlap risk because the manuscript draws heavily on the authors' prior Dart AOT work and on benchmark/task descriptions that may recur across related manuscripts. The references also signal multiple closely related papers in a short period. fileciteurlturn0file0	Even legitimate reuse can trigger reviewer concern if wording overlap is high or if dataset/task descriptions are recycled too closely.	Run iThenticate or equivalent, inspect self-overlap manually, and rewrite shared benchmark-description paragraphs into fresher wording.

One structural change would improve the manuscript more than any local copy-edit:



That reorganisation would align the argument more closely with what the paper already admits in its Threats to Validity section. fileciteurlturn0file0

Prioritised corrections

Severity	Correction	What to change concretely
Critical	Separate clean evaluation from harvested/transductive analysis	Create separate subsections and tables. Clean results should include only models not trained or repaired using the evaluation tasks. Coverage/harvest results should be clearly marked as optimisation analysis rather than generalisation evidence. filecite%turn0file0%1
Critical	Fix metric reporting inconsistency	Add compile@1 to Table 1 or remove the claim from Section 4. Define pass@k formally, state whether it is empirical k-sample success or an estimator, and cite the HumanEval convention. filecite%turn0file0%1 2
Critical	Add reproducibility package details	Include code/data/checkpoint availability, seeds, train/dev/test splits, sampling temperature/top-p, batch sizes, optimisation schedules, reward definitions, and hardware. filecite%turn0file0%1
High	Replace archive-facing run labels in main results	In the manuscript body and main tables, use descriptive names such as “Graph-conditioned, 16-prefix, no prompt assembly” rather than “G3”, and move internal IDs to an appendix. filecite%turn0file0%1
High	Normalise the references	Add venue/publisher/page range for all published items; add arXiv IDs or DOIs for preprints/technical reports; use consistent title capitalisation and punctuation. filecite%turn0file0%1 11
High	Add statistical uncertainty	Report confidence intervals for pass@k and compile@k, ideally with paired comparisons across tasks for major claims. filecite%turn0file0%1
Medium	Add Index Terms and venue-ready front matter	If non-blind: affiliations, emails, ORCIDs if required, keywords/index terms. If blind: remove identifying information consistently. IEEE expects Index Terms after the abstract. filecite%turn0file0%1 7
Medium	Tighten the prose	Remove colloquial register, shorten long sentences, and state interpretations with measured confidence. filecite%turn0file0%1
Medium	Improve captions and table clarity	Make captions standalone; expand abbreviations; mark best values; explain denominators in the table itself, not just in the caption. IEEE’s figure/table citation rules are a good benchmark if that is the target style. filecite%turn0file0%1 8
Medium	Add ethics / dual-use note	Because decompilation is relevant to malware analysis and reverse engineering, include a short dual-use and responsible-release statement. filecite%turn0file0%1

Annotated edits for key problematic passages

The examples below focus on passages where the draft is scientifically serviceable but stylistically or logically weaker than it needs to be.

Location	Original text	Issue	Suggested revision
Introduction	"This paper starts from that negative result."	Slightly conversational and backward-looking; it leans on prior work without immediately stating the present paper's contribution. filecite turn0file0	"Motivated by the finding that sequence-only fine-tuning does not reliably improve functional correctness, we evaluate whether graph-conditioned decoding, reward adaptation, reranking, and candidate pooling recover additional semantic fidelity."
Method	"The purpose is not to find one magical checkpoint but to map how each intervention changes the tradeoff..."	"One magical checkpoint" sounds colloquial and undercuts the otherwise technical tone. filecite turn0file0	"Our aim is not to identify a universally dominant checkpoint, but to characterise how each intervention shifts the trade-off among CodeBLEU, compilability, and functional correctness."
Results	"The strongest result appears when candidates are pooled across arms."	The sentence overstates the evidential status without immediately signalling that this is a pooled-coverage analysis, not a clean single-model win. filecite turn0file0	"The largest task-level coverage gain is observed when candidates are pooled across independently adapted arms, although this result reflects candidate-set complementarity rather than single-model generalisation."
Table/ Results narration	"The 17 files consist of the 14 standalone arms in Table 1 plus three additional repair/rerank pools..."	Archive-oriented detail interrupts the Results narrative. This belongs in an appendix or a dataset card for experiment artefacts. filecite turn0file0	"The multi-arm pool combines the 14 primary model families with three auxiliary repair/rerank pools; exact file identifiers are provided in the appendix for traceability."

A slightly larger paragraph-level rewrite is also warranted in the section on pooled candidates. The current paragraph is informative, but it mixes **headline result**, **pool composition**, **archive labels**, and **example tasks** in one block. A cleaner version would be:

"Pooling candidates across 17 evaluation pools increases task-level passing coverage from 24/154 for the clean graph-conditioned model and 28/154 for the best GRPO arm to 43/154 overall. This result indicates substantial complementarity across adaptation

strategies. Because some pooled arms include repair and harvest variants, we interpret this outcome as an upper-bound analysis of candidate discovery rather than as clean single-model generalisation. Full pool composition is reported in the appendix.”

filecite turn0file0

Suggested abstract and title options

Rewritten abstract

Neural decompilation is often evaluated with surface-form and compilability metrics, yet these measures can diverge from functional correctness. We study whether graph-conditioned decoding and inference-time candidate selection improve neural decompilation of Dart AOT binaries beyond sequence-only fine-tuning. Our approach encodes control-flow and lightweight data-flow structure extracted from x86-64 Dart AOT assembly using a GraphCodeBERT-based encoder and conditions a Qwen3-8B decoder through learned graph-prefix representations. We evaluate architecture ablations, GRPO-based reward adaptation, rejection-sampling supervised fine-tuning, compile-aware reranking, and multi-arm candidate pooling on 154 HumanEval-Dart tasks, with separate denominators for compile-oriented and pass-oriented evaluations. The clean graph-conditioned model provides the strongest overall balance of CodeBLEU, compilability, and functional correctness, reaching 0.638 CodeBLEU, compile@5 of 0.667, and pass@10 of 0.156. A binary-reward GRPO variant attains the highest standalone pass@10 of 0.182, but at substantial cost to compile@5 and CodeBLEU, indicating instability under sparse correctness rewards. No single adaptation arm dominates across all metrics. Instead, pooled candidate sets reveal substantial complementarity across arms, yielding at least one passing candidate for 43 of 154 tasks. We therefore conclude that graph conditioning improves the most balanced single-model profile, whereas multi-arm candidate generation and stronger selection mechanisms are the most promising route to higher functional recovery in Dart AOT decompilation.

filecite turn0file0

Improved title options

The current title is informative but dense. These alternatives are stronger because they reduce the subtitle’s list-like feel and foreground the paper’s actual claim.

Option	Why it is better
Graph-Conditioned Neural Decompilation of Dart AOT Binaries	Best if you want a concise, architecture-centred title.
Improving Functional Correctness in Dart AOT Neural Decompilation with Graph Conditioning and Candidate Ensembling	Best if you want the title to foreground the paper’s real outcome rather than the method inventory.
Graph Conditioning, Reward Adaptation, and Candidate Ensembling for Dart AOT Neural Decompilation	Best compromise if you want to keep the three-way method framing but shorten the current title.
Towards Functionally Correct Dart AOT Decompilation via Graph-Conditioned Generation and Candidate Pooling	Best if you want a more cautious, research-forward title.

My preferred option is the second one, because the paper’s most interesting story is not merely that it uses graph conditioning, but that it studies what actually helps **functional correctness**.

filecite turn0file0

Author checklist and recommended resources

Before resubmission, the authors should be able to answer **yes** to each item below.

Checklist item	Yes/No target
Have you separated clean held-out results from harvested or transductive optimisation results?	Yes
Have you defined pass@k precisely and made it comparable with prior code-generation work?	Yes
Have you restored or removed any promised-but-missing metrics such as compile@1?	Yes
Have you added confidence intervals or significance analyses for the main comparisons?	Yes
Have you replaced cryptic run labels in the main paper with descriptive names?	Yes
Have you added a code/data/checkpoint availability statement?	Yes
Have you reported seeds, sampling parameters, optimisation settings, and hardware?	Yes
Have you normalised all references and added missing persistent identifiers?	Yes
Have you added keywords/index terms and venue-correct front matter?	Yes
Have you run a similarity check for plagiarism and self-overlap?	Yes
Have you added an ethics or dual-use statement appropriate to decompilation research?	Yes
Have you completed one full venue-specific style pass after choosing the target outlet?	Yes

The most useful resources to consult next are these:

- **IEEE Editorial Style Manual for Authors.** This is the best primary source if the manuscript remains in numbered-citation technical-paper form; it specifies article order, one-paragraph abstracts, Index Terms, heading conventions, reference handling, and figure/table citation practice. ¹²
- **APA Publication Manual and APA Style guidance.** Use this only if the target outlet truly requires APA, because it would require a full conversion to author-date citations and APA-specific paper conventions. ⁵
- **HumanEval / “Evaluating Large Language Models Trained on Code”.** Use this as the reference point for explaining pass@k clearly and avoiding metric ambiguity. ²
- **Crossref / DOI display guidance.** Use actionable DOI or stable identifier formatting consistently wherever DOIs are available. ¹³

- **Reference metadata for omitted identifiers.** Several cited works in the paper have readily available persistent records or arXiv identifiers, including GraphCodeBERT, LoRA, DeepSeekMath, SimKO, HumanEval/Codex, and the Qwen3 Technical Report. ¹⁴

On substance rather than style, the paper’s revised version should aim to make one claim unmistakably clear: **graph conditioning improves the clean single-model baseline, while pooled candidate coverage is a separate and more exploratory result about complementarity and selection.** If the revision makes that distinction crisp, tightens the metrics, and strengthens reproducibility, the paper will read as a much more rigorous and credible contribution. turn0file0

¹ ⁴ ⁶ ⁷ ⁸ ¹⁰ ¹² IEEE Editorial Style Manual for Authors

<https://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/7/IEEE-Editorial-Style-Manual-for-Authors.pdf>

² Evaluating Large Language Models Trained on Code

https://arxiv.org/abs/2107.03374?utm_source=chatgpt.com

³ ⁹ ¹¹ ¹⁴ GraphCodeBERT: Pre-training Code Representations with Data Flow

https://arxiv.org/abs/2009.08366?utm_source=chatgpt.com

⁵ APA style - Wikipedia

https://en.wikipedia.org/wiki/APA_style

¹³ Digital object identifier

https://en.wikipedia.org/wiki/Digital_object_identifier?utm_source=chatgpt.com